Lab-2: Excel Functions (Theory + Formula)

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An Excel function is a ready-made formula: `=FUNCTION_NAME(argument1, argument2, ...)`. This lab covers the functions taught in Class 1 (the Functions and SUMIF-SUMIFS sheets), each with its theory, formula and the result for the class sample data.

1. Basic Calculation Functions: SUM, PRODUCT, AVERAGE, MIN, MAX

Sample data: students Deepa 85, Binil 96, suman 78, Shilpa 65, Sunil 89 in B4:B8. Products P1 to P5 have quantity in E4:E8 (150, 200, 250, 300, 350) and cost in F4:F8 (10, 15, 20, 25, 30).

Function	Theory	Formula	Result
SUM	Adds all numbers in a range	`=SUM(B4:B8)`	413
PRODUCT	Multiplies numbers; used here for Total = Qty × Cost	`=PRODUCT(E4,F4)`	1500 (P1)
AVERAGE	Adds the numbers and divides by how many there are	`=AVERAGE(J4:J8)`	82.6
MIN	Returns the smallest number in a range	`=MIN(M4:M8)`	1 (marks 85, 1, 8, 65, 89)
MAX	Returns the largest number in a range	`=MAX(B15:B19)`	96

2. Counting Functions: COUNT, COUNTA, COUNTBLANK, COUNTIF, COUNTIFS

COUNT counts only numbers, COUNTA counts any non-empty cell, COUNTBLANK counts empty cells. COUNTIF counts cells that meet one condition; COUNTIFS counts rows that meet several conditions at once.

Function	Theory	Formula	Result
COUNT	Counts cells that contain numbers	`=COUNT(E15:E19)`	5
COUNTA	Counts cells that are not empty (text or numbers)	`=COUNTA(H15:H19)`	5 (A, B, C, D and 85)
COUNTBLANK	Counts empty cells	`=COUNTBLANK(K15:K19)`	1 (K16 is empty)
COUNTIF	Counts cells meeting a single condition	`=COUNTIF(E26:E30,">50")`	3 (marks 55, 60, 90)
COUNTIFS	Counts rows meeting multiple conditions	`=COUNTIFS(G25:G29,"IT",H25:H29,">35000")`	2

Class practice questions

Question	Formula	Result
No. of female students (B26:B30 = F, F, M, F, M)	`=COUNTIF(B26:B30,"F")`	3
No. of male students	`=COUNTIF(B26:B30,"M")`	2
Students below 50 marks (E26:E30 = 55, 60, 30, 90, 44)	`=COUNTIF(E26:E30,"<50")`	2
Q1: Count IT employees with salary above 35000	`=COUNTIFS(G25:G29,"IT",H25:H29,">35000")`	2 (40000, 55000)
Q2: Count HR employees with salary exactly 60000	`=COUNTIFS(G25:G29,"HR",H25:H29,60000)`	1

3. Logical Functions: IF and IFS

IF returns one value when a condition is true and another when it is false. IFS checks several conditions in order and returns the value for the first one that is true, so it replaces a long nested IF.

IF: Pass or Fail. Rule: marks of 40 or above show Pass, otherwise Fail. Formula: `=IF(B41>=40,"PASS","FAIL")`. Note that the class sheet uses `>40`, which would wrongly fail Sneha's 40, so `>=40` is the correct form for this rule.

Student	Marks	Result
Amit	85	PASS
Priya	35	FAIL
Rahul	65	PASS
Sneha	40	PASS
Rohan	25	FAIL
Neha	90	PASS

IFS: Grades. Rules: 90 and above A+, 80 to 89 A, 70 to 79 B, 60 to 69 C, below 60 Fail.

```
=IFS(B51>=90,"A+", B51>=80,"A", B51>=70,"B", B51>=60,"C", B51<60,"Fail")
```

Student	Marks	Grade
Amit	95	A+
Priya	82	A
Rahul	76	B
Sneha	68	C
Rohan	55	Fail
Neha	91	A+

Practice Q3: Bonus eligibility (IF). Sales of 60000 or more is Eligible, otherwise Not Eligible: `=IF(B63>=60000,"Eligible","Not Eligible")`.

Employee	Sales	Bonus
Rahul	45000	Not Eligible
Sneha	65000	Eligible
Ankit	90000	Eligible
Riya	30000	Not Eligible
Mohit	75000	Eligible

Practice Q4: Performance rating (IFS). 90 to 100 Excellent, 80 to 89 Very Good, 70 to 79 Good, 60 to 69 Average, below 60 Poor.

```
=IFS(B72>=90,"Excellent", B72>=80,"Very Good", B72>=70,"Good",
B72>=60,"Average", B72<60,"Poor")
```

Employee	Score	Performance
Rahul	98	Excellent
Sneha	84	Very Good
Ankit	76	Good
Riya	68	Average
Mohit	50	Poor

4. Conditional Sum Functions: SUMIF and SUMIFS

SUMIF adds the values that meet one condition. SUMIFS adds the values that meet several conditions together. Syntax: `=SUMIF(criteria_range, criteria, sum_range)` and `=SUMIFS(sum_range, criteria_range1, criteria1, criteria_range2, criteria2)`.

SUMIF example. Total score by Pokemon type, with Type in E6:E14, Total in F6:F14 and the type to match in H7:H9: `=SUMIF(E6:E14,H7,F6:F14)`.

Type	Formula	Result
Grass	`=SUMIF(E6:E14,"Grass",F6:F14)`	1248 (318 + 405 + 525)
Fire	`=SUMIF(E6:E14,"Fire",F6:F14)`	1248 (309 + 405 + 534)

Type	Formula	Result
Water	`=SUMIF(E6:E14,"Water",F6:F14)`	1249 (314 + 405 + 530)

SUMIFS example. Employee data: Department in E24:E31, Status in F24:F31, Salary in G24:G31.

Question	Formula	Result
Q1: Total salary of employees in IT and Present	`=SUMIFS(G24:G31,E24:E31,"IT",F24:F31,"Present")`	77000 (Amit 35000 + Riya 42000)
Q2: Total salary of employees in HR and Present	`=SUMIFS(G24:G31,E24:E31,"HR",F24:F31,"Present")`	90000 (Neha 28000 + Priya 30000 + Sneha 32000)

5. Helper Functions: ISNUMBER, CHAR, ROW

Function	Theory	Formula	Result
ISNUMBER	Returns TRUE if the cell holds a number, otherwise FALSE	`=ISNUMBER(B30)`	TRUE for 500, FALSE for text or an empty cell
ROW	Returns the row number of a cell reference	`=ROW(A3)`	3
CHAR	Returns the character for a code number (65 = A)	`=CHAR(65)`	A
CHAR + ROW	Fills letters down a column automatically	`=CHAR(64+ROW(A1))`	A, then B, C, D as the formula is copied down
IF + ISNUMBER	Running balance that stays blank on empty rows	`=IF(ISNUMBER(B30),G29+C30-E30,"")`	Previous balance + credit − debit

Conclusion

SUM, PRODUCT, AVERAGE, MIN and MAX do the basic calculations; the COUNT family counts data; IF and IFS make decisions; SUMIF and SUMIFS add values by condition. Together they cover most everyday data tasks in Excel.

Source: Class 1 workbook, sheets Functions and SUMIF-SUMIFS-FUNCTION. Results were worked out by hand from the sample data, not run in Excel.